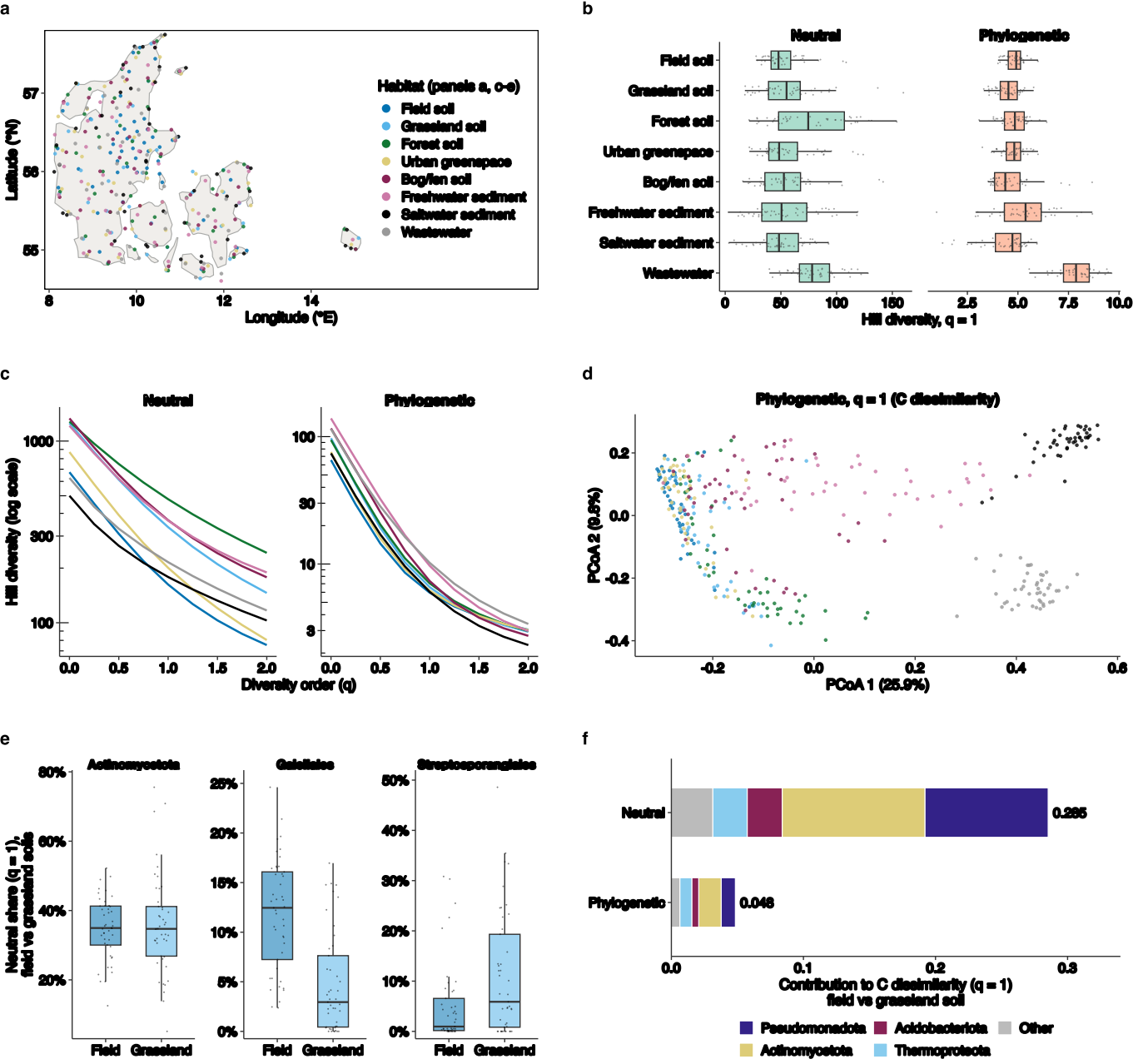

a
Habitat (panels a, c–e)
Field soil
Grassland soil
Forest soil
Urban greenspace
Bog/fen soil
Freshwater sediment
Saltwater sediment
Wastewater
Latitude (°N)
Longitude (°E)
b
Neutral
Phylogenetic
Hill diversity, q = 1
c
Neutral
Phylogenetic
Hill diversity (log scale)
Diversity order (q)
d
Phylogenetic, q = 1 (C dissimilarity)
PCoA 2 (9.6%)
PCoA 1 (25.9%)
e
Actinomycetota
Gaiellales
Streptosporangiales
Neutral share (q = 1), field vs grassland soils
Field
Grassland
f
Neutral
Phylogenetic
0.265
0.048
Contribution to C dissimilarity (q = 1) field vs grassland soil
Pseudomonadota
Acidobacteriota
Other
Actinomycetota
Thermoproteota